Causality via Vector Autogressive Models and Structural Equation Models

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[THIS LECTURE IS INCOMPLETE. WILL BE COMPLETED SOON.]

1 Introduction

Causality is a fundamental concept in science, aiming to determine whether one event influences another. In time series analysis, **Granger causality**, introduced by Clive Granger in 1969, provides a statistical framework for assessing whether past values of one time series help predict another. Unlike traditional notions of causality in physics or philosophy, Granger causality is based on predictability if including past values of one time series improves the forecast of another. Granger causality has wide applications in economics, neuroscience, climate science, and engineering. In a *multivariate time series*, where multiple variables evolve over time, Granger causality extends naturally to examine interactions among multiple time-dependent signals. This is most often used causal model in brain network analysis. Granger causality is usually built on top of Vector Autoregressive (VAR) models.

2 Vector Autoregressive (VAR) Model

The **Vector Autoregressive (VAR)** model is a fundamental framework for modeling multivariate time series, where multiple variables evolve over time and potentially influence each other. Unlike univariate autoregressive models that describe the evolution of a single time series, the VAR model extends this concept to a system of interdependent time series. This makes it particularly useful for capturing dynamic relationships in fields such as economics, neuroscience, and climate science.

Consider a system of N time series denoted as

$$X(t) = [X_1(t), X_2(t), \dots, X_N(t)]^\top,$$

where $X_i(t)$ represents the time series of the i-th region at time t. The VAR model of order P, denoted as VAR(P), expresses each time series as a linear

combination of its own past values and the past values of all other variables in the system. $X(t) \in \mathbb{R}^N$ is a time-varying vector. This model is formulated as

$$X(t) = \sum_{p=1}^{P} A_p X(t-p) + \varepsilon(t), \tag{1}$$

where A_p are $N \times N$ coefficient matrices that capture the influence of past values at lag p, and $\varepsilon(t)$ is an N-dimensional vector of error terms, assumed to be normally distributed with zero mean and a covariance matrix Σ, i.e.,

$$\varepsilon(t) \sim \mathcal{N}(0, \Sigma).$$

The number of lags P is a crucial hyperparameter that determines how far back in time the model considers past information. Expanding each component, we can write

$$X_i(t) = \sum_{p=1}^{P} \sum_{j=1}^{N} A_{ij}^{(p)} X_j(t-p) + \varepsilon_i(t), \tag{2}$$

where $A_{ij}^{(p)}$ represents the effect of the j-th variable at lag p on the i-th variable. Note $A_p = (A_{ij}^{(p)})$. If $A_{ij}^{(p)}$ is significantly different from zero, this indicates a directional relationship where past values of $X_j(t)$ contribute to predicting $X_i(t)$, which is the foundation for testing Granger causality.

The estimation of VAR model parameters is typically performed using least squares estimation:

$$\min_{A_1, \cdots, A_P} \sum_{t=P+1}^{T} \|X(t) - \sum_{p=1}^{P} A_p X(t-p)\|^2. \tag{3}$$

Instead of estimating the coefficients iteratively, which will compound errors, we can estimate the entire coefficient matrix simultaneously in a single step by reformulating the problem in matrix form. Let

$$A = [A_1, A_2, \ldots, A_P] \in \mathbb{R}^{N \times (NP)}, \tag{4}$$

which stacks all lag-specific coefficient matrices horizontally. Next, define the regressor matrix:

$$X_{\text{reg}}(t) = [X(t-1)^\top, X(t-2)^\top, \ldots, X(t-P)^\top]^\top \in \mathbb{R}^{(NP) \times 1}. \tag{5}$$

This allows the VAR model to be rewritten in a compact form as

$$X(t) = A X_{\text{reg}}(t) + \varepsilon(t), \quad t = P+1, P+2, \cdots T \tag{6}$$

where $\varepsilon(t)$ is the residual error term assumed to follow a multivariate normal distribution with zero mean and covariance Σ, i.e.,

$$\varepsilon(t) \sim \mathcal{N}(0, \Sigma). \tag{7}$$

Stacking the equations over all time points from $P+1$ to T, we define

$$Y = [X(P+1), X(P+2), \ldots, X(T)] \in \mathbb{R}^{N\times(T-P)}, \tag{8}$$

where Y represents the matrix of observed time series from $P+1$ to T. Similarly we define

$$\mathbf{X}_{\text{reg}} = [X_{\text{reg}}(P+1), X_{\text{reg}}(P+2), \ldots, X_{\text{reg}}(T)] \in \mathbb{R}^{(NP)\times(T-P)}, \tag{9}$$

Then we can write the VAR as a single matrix equation as

$$Y = A\mathbf{X}_{\text{reg}} + E, \tag{10}$$

where E is the residual error matrix

$$E = [\varepsilon(P+1), \varepsilon(P+2), \ldots, \varepsilon(T)] \in \mathbb{R}^{N\times(T-P)}. \tag{11}$$

To estimate A, we minimize the sum of squared residuals:

$$\widehat{A} = \arg\min_{A} \|Y - A\mathbf{X}_{\text{reg}}\|_F^2, \tag{12}$$

where $\|\cdot\|_F$ denotes the Frobenius norm. The least squares solution is given by

$$\widehat{A} = Y\mathbf{X}_{\text{reg}}^{\top}\left(\mathbf{X}_{\text{reg}}\mathbf{X}_{\text{reg}}^{\top}\right)^{-1}. \tag{13}$$

Once the coefficient matrix $\widehat{A}$ is obtained, the predicted values for all time points can be computed as

$$X_{\text{pred}} = \widehat{A}\mathbf{X}_{\text{reg}}. \tag{14}$$

This formulation ensures that all parameters are estimated simultaneously in a single step, minimizing the total squared error over all time points. The simultaneous estimation method provides a more stable and efficient approach compared to iterative forecasting methods, particularly when dealing with high-dimensional time series data.

The Vector Autoregressive (VAR) model requires past values of the time series $X(1), \ldots, X(P)$ to make predictions. The selection of the optimal lag order P is an essential step in model fitting. Choosing too small a value may result in missing important dependencies, whereas an excessively large P may introduce overfitting. The lag order is typically determined using information criteria such as the Akaike Information Criterion (AIC) or the Bayesian Information Criterion (BIC), which balance model complexity with goodness of fit.

Despite its flexibility and interpretability, the VAR model assumes stationarity and its statistical properties of the time series remain constant over time. Additionally, VAR models assume linear relationships, which may not fully capture nonlinear dependencies present in brain network data.

The VAR framework provides a powerful foundation for analyzing multivariate time series and forms the basis for testing Granger causality. By capturing interactions among multiple variables, it serves as an essential tool for investigating directional dependencies and understanding the underlying structure of complex dynamic systems.

3 Granger Causality

Granger causality provides a statistical framework for determining whether the past values of one time series improve the prediction of another within a multivariate time series setting. Unlike traditional notions of causality, which imply a mechanistic link, Granger causality is based purely on predictive ability: if including past values of a variable $X_j(t)$ significantly reduces the prediction error of another variable $X_i(t)$, then $X_j(t)$ is said to **Granger-cause** $X_i(t)$.

Given a multivariate system modeled by a vector autoregressive process of order P (VAR(P)):

$$X_i(t) = \sum_{p=1}^{P} \sum_{j=1}^{N} A_{ij}^{(p)} X_j(t-p) + \varepsilon_i(t), \tag{15}$$

where $A_{ij}^{(p)}$ are the autoregressive coefficients capturing the influence of variable j on variable i at lag p, and $\varepsilon_i(t)$ is a zero-mean white noise process with covariance Σ. To test whether $X_j(t)$ Granger-causes $X_i(t)$, we compare two models:

- **Full model:** The standard VAR(P) equation, where $X_i(t)$ depends on the past values of all variables, including $X_j(t-p)$, given by:

$$X_i(t) = \sum_{p=1}^{P} \sum_{j=1}^{N} A_{ij}^{(p)} X_j(t-p) + \varepsilon_i(t). \tag{16}$$

- **Restricted model:** A modified VAR(P) equation where the past values of $X_j(t)$ are excluded from the predictors:

$$X_i(t) = \sum_{p=1}^{P} \sum_{l \neq j} A_{il}^{(p)} X_l(t-p) + \varepsilon_i'(t). \tag{17}$$

We then determine whether the exclusion of $X_j(t-p)$ significantly worsens the prediction of $X_i(t)$. If the residual error in the restricted model increases significantly compared to the full model, we conclude that $X_j(t)$ provides useful predictive information for $X_i(t)$ and therefore Granger-causes it. To statistically test this, we define the hypotheses as follows:

$$H_0 : A_{ij}^{(p)} = 0, \quad \text{for all } p \in \{1, \ldots, P\}, \tag{18}$$

$$H_A : A_{ij}^{(p)} \neq 0, \quad \text{for at least one } p. \tag{19}$$

Under the null hypothesis H_0, the past values of $X_j(t)$ do not contribute to predicting $X_i(t)$, implying no Granger causality. If the null hypothesis is rejected, it suggests that past values of $X_j(t)$ improve the prediction

of $X_i(t)$, supporting the presence of Granger causality. Thus, we use the F-statistic computed as

$$F = \frac{\big(\mathrm{RSS}_{\mathrm{restricted}} - \mathrm{RSS}_{\mathrm{full}}\big)/P}{\mathrm{RSS}_{\mathrm{full}}/\big(T - P - NP\big)} \sim F_{P,\,T-P-NP}, \tag{20}$$

where: $\mathrm{RSS}_{\mathrm{restricted}}$ is the residual sum of squares for the restricted model, $\mathrm{RSS}_{\mathrm{full}}$ is the residual sum of squares for the full model, and NP is the number of regressors in the full model for one equation.

Under H_0, the corresponding p-value is computed as

$$p = 1 - F_{P,\,T-P-NP}(F), \tag{21}$$

with the cumulative distribution function $F_{P,\,T-P-NP}$.

3.1 Granger Causality Matrix

For large numbers of regions (N), the test results are often summarized in a *Granger causality matrix* $G = (G_{ij})$, where each entry G_{ij} is the p-value of the test for whether the past values of $X_j(t)$ significantly improve the prediction of $X_i(t)$. Smaller p-values indicate stronger evidence of Granger causality. Thus, we can construct a weighted directed graph $W = (w_{ij})$ by defining

$$w_{ij} = 1 - G_{ij},$$

so that larger values of w_{ij} correspond to stronger causal influence from X_j to X_i. In this representation, the edge weights directly reflect the strength of the detected Granger causal relationships. This provides a convenient way to visualize directional relationships among brain regions. In neuroscience, Granger causality is widely used to infer functional connectivity from fMRI and EEG data, helping to reveal how activity propagates across brain networks. Beyond neuroscience, it is also a standard tool in econometrics, where it is applied to examine predictive relations among financial indicators such as interest rates, exchange rates, and stock prices.

4 Structural Equation Model

While Granger causality provides a predictive notion of temporal influence by testing whether past values of one variable improve the forecast of another, an alternative and complementary approach is provided by the *Structural Equation Model (SEM)* framework. SEM is rooted in econometrics and psychometrics, and has been widely adopted in neuroscience for analyzing brain networks from fMRI and EEG data. Unlike Granger causality, which is inherently temporal and based on lagged dependencies, SEM focuses on *contemporaneous relationships* among variables at

the same time point. In SEM, directed edges represent direct causal effects among observed variables, and the model is built as a system of simultaneous linear equations.

4.1 Structural Equation Model Formulation

Consider a system of N observed variables at time t, denoted by

$$X(t) = [X_1(t), X_2(t), \ldots, X_N(t)]^\top \in \mathbb{R}^N.$$

The SEM formulation assumes that each variable $X_i(t)$ can be expressed as a linear combination of the other variables in the system plus a residual noise term:

$$X_i(t) = \sum_{j \neq i} B_{ij} X_j(t) + \varepsilon_i(t), \tag{22}$$

where B_{ij} denotes the direct influence of variable j on variable i, and $\varepsilon_i(t)$ is an independent noise term. Stacking all N equations, we obtain the simultaneous equations system

$$X(t) = BX(t) + \varepsilon(t), \tag{23}$$

where $B \in \mathbb{R}^{N \times N}$ is the coefficient matrix with zeros along the diagonal (no self-influence), and $\varepsilon(t) \sim \mathcal{N}(0, \Psi)$ is a vector of residual errors with covariance matrix Ψ. Rearranging,

$$(I - B)X(t) = \varepsilon(t). \tag{24}$$

The coefficient B_{ij} quantifies the instantaneous causal influence from X_j to X_i. SEM assumes a <u>recursive system without algebraic loops</u>.

4.2 Estimation of Model Parameters

Given data $X = [X(1), X(2), \ldots, X(T)] \in \mathbb{R}^{N \times T}$ across T time points, the SEM coefficients are typically estimated by ordinary least squares (OLS) regression performed for each variable X_i against the others:

$$\widehat{B}_{i,-i} = \arg \min_{\beta \in \mathbb{R}^{N-1}} \|X_i - \beta^\top X_{-i}\|^2, \tag{25}$$

where $X_i = [X_i(1), \ldots, X_i(T)]^\top$ is the time series of the i-th variable and X_{-i} is the $(T \times (N-1))$ matrix of all other variables. The residual vector $\widehat{\varepsilon}_i$ is then computed as

$$\widehat{\varepsilon}_i = X_i - \widehat{B}_{i,-i}^\top X_{-i}.$$

Stacking across all i, we obtain the estimated coefficient matrix $\widehat{B}$ and residuals $\widehat{\varepsilon}$.

This procedure can also be written in compact form as

$$\widehat{B} = \arg\min_{B} \|X - BX\|_F^2, \tag{26}$$

subject to $\mathrm{diag}(B) = 0$, where $\|\cdot\|_F$ is the Frobenius norm. The solution can be computed efficiently using pseudo-inverse operations and is robust to high-dimensional settings.

4.3 SEM-based Causality Testing

To test whether brain region X_j has a causal effect on X_i, we examine the null hypothesis

$$H_0 : B_{ij} = 0, \tag{27}$$

$$H_A : B_{ij} \neq 0. \tag{28}$$

The test is performed using a t-test on the regression coefficient $\widehat{B}_{ij}$, where the standard error is obtained from the OLS residual variance and the covariance of the design matrix:

$$\mathrm{se}(\widehat{B}_{ij}) = \sqrt{\widehat{\sigma}^2 \left[(X_{-i}^\top X_{-i})^{-1}\right]_{jj}}.$$

The test statistic is then

$$t_{ij} = \frac{\widehat{B}_{ij}}{\mathrm{se}(\widehat{B}_{ij})},$$

with corresponding p-value

$$p_{ij} = 2\big(1 - F_{t,\,T-N}(|t_{ij}|)\big),$$

where $F_{t,\,T-N}$ is the cumulative distribution of a t-distribution with $T-N$ degrees of freedom. The collection of all p_{ij} values forms the *SEM causality matrix*.

4.4 SEM Causality Matrix

Analogous to the Granger causality matrix, SEM results can be summarized in a matrix $S = (S_{ij})$:

$$S_{ij} = p\text{-value of test for } B_{ij} = 0, \tag{29}$$

where each entry represents the statistical evidence for an instantaneous causal effect from X_j to X_i. Smaller values of S_{ij} indicate stronger evidence of causality. A weighted directed graph can then be constructed using

$$w_{ij} = 1 - S_{ij},$$

so that larger weights correspond to stronger contemporaneous causal relationships. This visualization highlights the immediate directional influences among nodes, in contrast to Granger causality which is based on lagged dependencies.

Table 1: Comparison of Structural Equation Modeling (SEM) and Granger Causality (GC).

Aspect	**Granger Causality**	**SEM**
Temporal vs. Instantaneous	Captures *lagged* influences; past values improve prediction.	Quantifies *instantaneous* interactions at the same time point.
Modeling framework	Built on Vector Autoregressive (VAR) models with lag order P.	Uses simultaneous regression equations with coefficient matrix B and zero diagonals.
Interpretation	Reflects directed *information flow* over time.	Reflects *direct coupling* or contemporaneous effects.
Statistical testing	Tests $H_0 : A_{ij}^{(p)} = 0$; uses F-tests on full vs. restricted VAR models.	Tests $H_0 : B_{ij} = 0$; uses t-tests on regression coefficients.
Graph representation	GC matrix $G = (G_{ij})$; edges encode temporal causality.	SEM matrix $S = (S_{ij})$; edges encode instantaneous causality.

4.5 Comparison Granger and SEM Causality

SEM and Granger causality provide complementary views of causality in multivariate systems. Table 1 highlights their main differences. Both approaches provide valuable but distinct perspectives. Granger causality is inherently temporal, detecting whether one variable's past improves another's forecast, making it suitable for studying information flow across time. In contrast, SEM emphasizes contemporaneous dependencies, identifying direct instantaneous couplings between variables. This distinction is particularly important in neuroscience: SEM can highlight structural or effective connectivity at a single time point, while Granger causality uncovers dynamic propagation of activity across brain regions. Using both frameworks in tandem can thus provide a more complete characterization of causal interactions in complex systems.

4.6 Limitation of Granger and SEM Causality

Granger causality and SEM are primarily designed to estimate local, pairwise directional relationships through regression-based frameworks. They rely on assumptions such as linearity, sufficient sample size, and a well-conditioned covariance structure. As the number of variables increases, the number of parameters grows rapidly, making estimation increasingly sensitive to noise, model specification, and multicollinearity. As a result, stable recovery of the underlying network structure becomes difficult in high-dimensional settings.

In principle, if a true adjacency matrix were known, one could apply graph-theoretic algorithms (e.g., Johnson's or Tarjan's) to enumerate cy-

cles. However, in practice the adjacency must be estimated from data, and this estimation step is precisely where these methods break down.

When the data are noisy, as is typical in applications such as fMRI, the estimated adjacency matrices from Granger causality or SEM tend to become dense, with many small but nonzero coefficients. Noise inflates spurious correlations, making it difficult to distinguish true causal effects from random fluctuations, so inferred cycles often reflect noise rather than meaningful structure.

This issue is further amplified in large-scale systems. In high-dimensional settings with many interacting variables, over-parameterization and multicollinearity lead to unstable estimates, where many different models fit the data equally well. Consequently, reliable identification of cyclic structure becomes ill-posed, as there is no clear way to separate true feedback loops from transient or noise-driven interactions.